

StrumCoach

Real-time guitar strumming feedback on Wear OS and Android,
via spectral-flux onset detection

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Task and motivation

Rhythm guitar practice is self-corrected. Metronomes and tab players tell a beginner what to play, but not whether it was actually played in time and in the right direction.

Detecting single strokes is hard for two reasons:

- **Motion Over-Counts:** follow-through and return swing move the wrist without the pick touching a string.
- **Audio Masking:** fast patterns pack strokes within a few hundred milliseconds, and a quiet strike arrives while a louder note is still decaying.

Wrist motion alone



Loudness alone



Why two modalities

Audio

The only signal that confirms the strings were actually struck. A picked string is a broadband transient across many frequency bins.

Blind to direction

Wrist motion

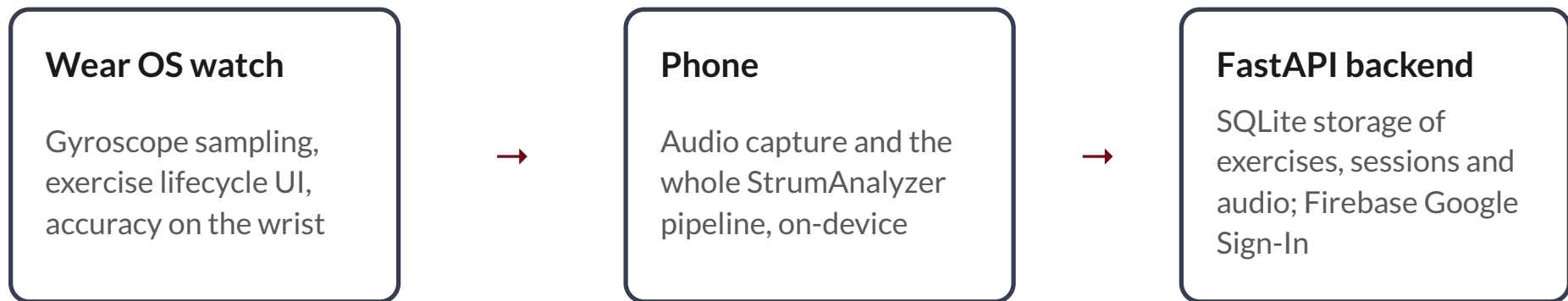
A gyroscope on the strumming wrist separates a downstroke from an upstroke by the sign of its rotation.

Cannot tell a strike from a silent return swing

Fusion: audio decides *when* a stroke happened, motion decides *which way* it went and vetoes sounds with no real swing.



System architecture



- **Same Window:** the phone records while the watch is in the execution phase, so both streams cover the same wall-clock window.
- **One Index Space:** two clocks have no reliable offset, so envelope and flux are resampled onto the sensor sample index before detection runs.



Spectral flux - the idea

- **A Fingerprint Every 10 ms:** each frame is a bar chart of how much energy sits in every frequency.
- **Flux Is The Change:** compare one fingerprint to the previous one and add up only the bars that grew.
- **A Pick Hitting Strings:** many bars grow at once, so the sum spikes.
- **A Note Fading Out:** the same bars, only smaller, so nothing grew and the sum stays flat.

Flux asks whether the sound *changed shape*, not whether it got louder.

frame n-1 — the note ringing



frame n — a new strike



what grew — summed, this is the flux

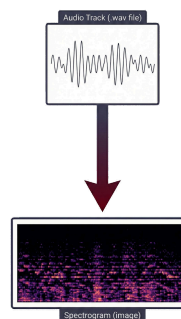


Spectral flux extraction

- **STFT:** a 25 ms window frame every 10 ms hop, zero-padded to the next power of two.
- **FFT:** an iterative, in-place radix-2 Cooley-Tukey transform written in Kotlin, with no external DSP library.
- **Normalization:** the flux series is scaled to $[0, 1]$ by its own session maximum, so thresholds hold regardless of recording volume.

$$SF[n] = \sum_k \max(0, M_n[k] - M_{n-1}[k])$$

Half-wave rectified frame-to-frame difference of the magnitude spectrum, summed across bins.



The masked strike

A quiet but genuine strike can arrive while a louder preceding note is still audibly decaying. Total energy never rises, so amplitude and RMS novelty miss it; the wrist signal alone cannot confirm it either.

Spectral flux recovers it: a picked string briefly excites many bins at once, while a decaying tail is narrowband and smooth. Flux reacts to the change in spectral shape, not to loudness.

Amplitude envelope

second strike hidden



Spectral flux

still a clear spike

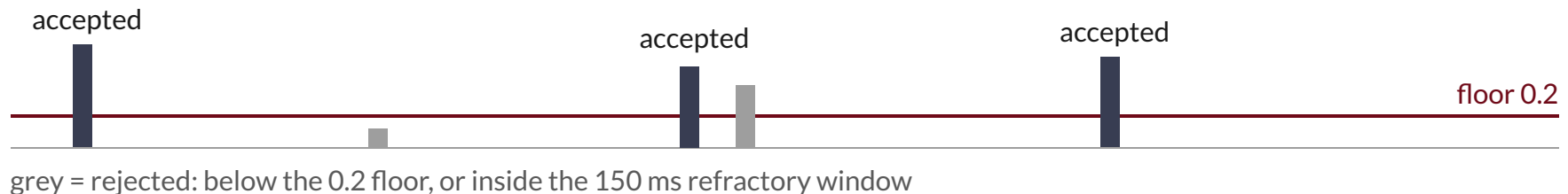


Onset candidate selection

Flux-based (preferred)

$$SF[n] \geq 0.2 \wedge SF[n] - \min_{i \in [n-8, n)} SF[i] \geq 0.1$$

A peak must clear an absolute floor, stand out from its trailing minimum, and respect a 150 ms refractory period. Absolute thresholds, not fractions of the loudest strike.



Gyro-only (fallback)

Only when no flux could be extracted. Samples are grouped into runs of constant sign; the strongest sample per run above a fixed threshold is kept, spaced by 120 ms.

Cannot separate a genuine stroke from a silent return swing, so it is materially less reliable.

Motion gate: an audio-confirmed candidate also needs gyroscope magnitude ≥ 0.4 nearby — a loud sound with no wrist swing is rejected.

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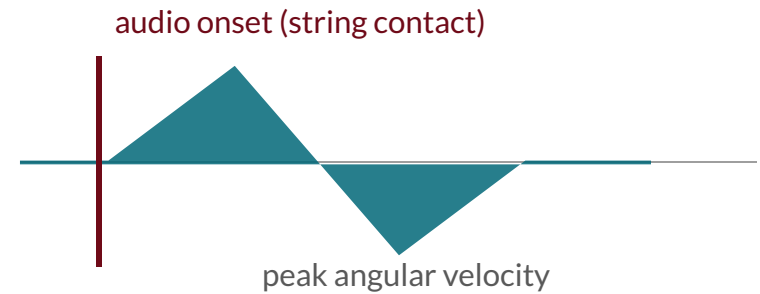
Direction classification

The gyroscope z-axis is the only motion input. A single-pole low-pass filter removes sensor jitter before any peak detection:

$$y[n] = y[n-1] + \alpha (x[n] - y[n-1]), \quad \alpha = 0.5$$

Each accepted onset is then labelled from the **sign of the gyroscope z-value** in the direction-search window: positive is one stroke direction, negative the other.

Why search forward



↓ **D** | ↑ **U**
 $z > 0$ | $z < 0$

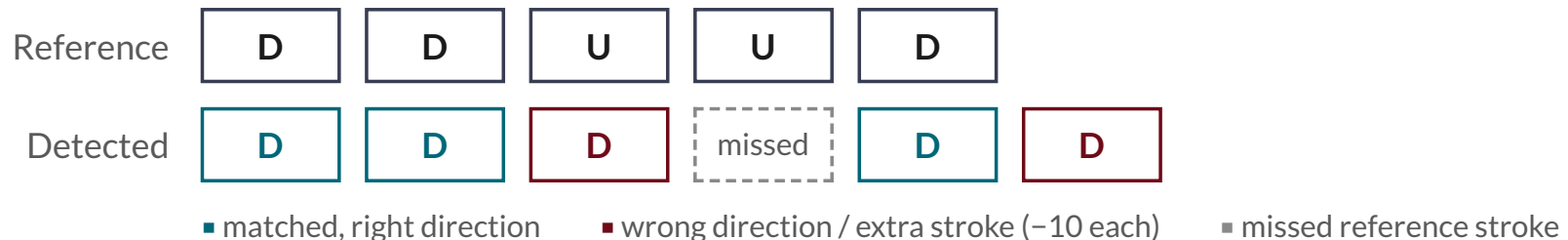
one sign,
one direction



Accuracy scoring

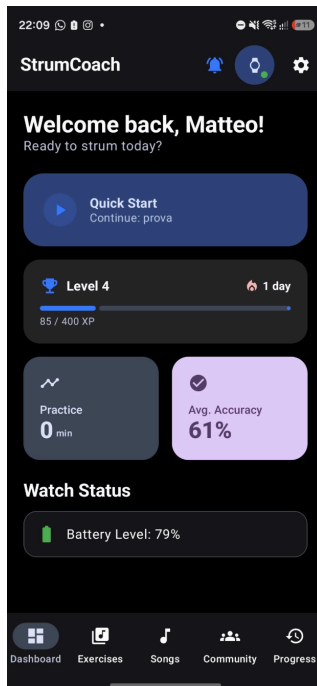
Detected strokes are aligned to the reference by the offset between the two first strokes, then each is greedily matched to its nearest unclaimed reference stroke within a 15-sample window. A reference stroke can be claimed only once, so over-strumming cannot inflate the score.

$$\text{Acc} = \text{clamp}_{[0,100]} \left(\frac{\text{matched, correct-direction strokes}}{|\text{reference strokes}|} \times 100 - 15 \cdot |\text{timing error}| - 10 \cdot n_{\text{extra}} \right)$$

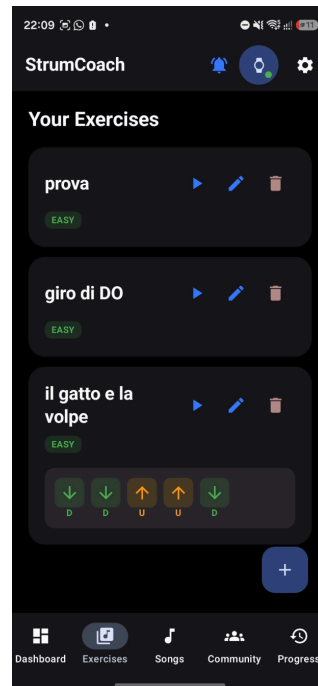


The phone app

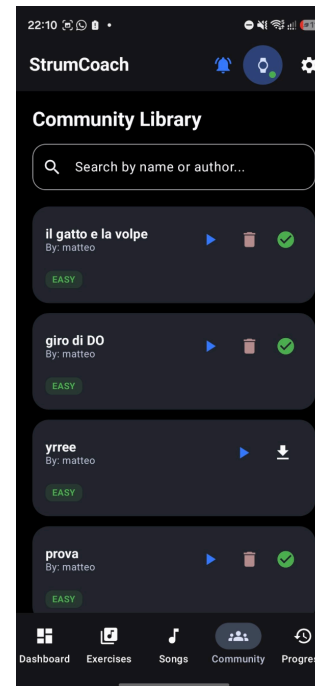
Jetpack Compose and Material 3, gated behind Firebase Google Sign-In.



Dashboard



Exercises



Community

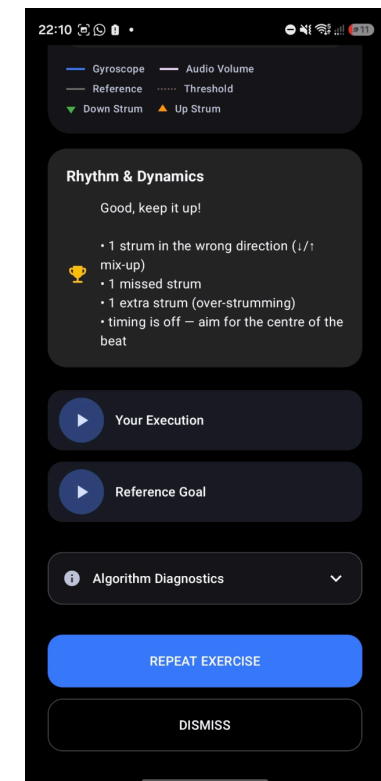
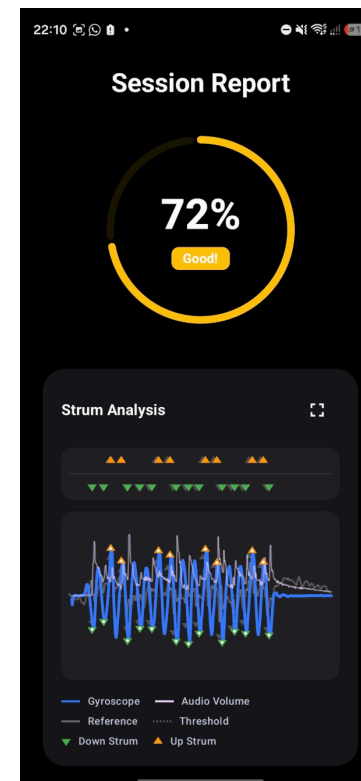


Progress

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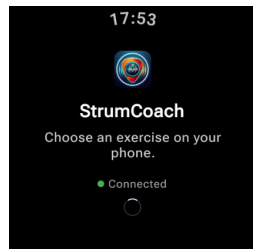
Session report

- **Accuracy Ring:** the score of accuracy with a qualitative grade.
- **Strum Analysis:** the detected down/up sequence compared to the reference, beat by beat.
- **Waveform:** the gyroscope trace with detected strums and, when available, the audio-onset threshold overlaid.
- **Rhythm and Dynamics:** the errors named in words — wrong directions, missed and extra strums, timing drift.



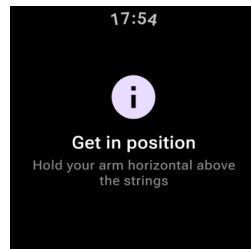
On the wrist

The watch mirrors the session lifecycle driven by the phone, and shows the result without the user having to pick the phone up.



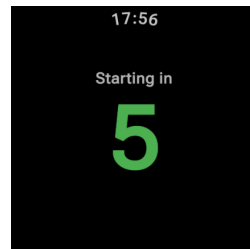
Idle

Connection status



Ready check

Arm horizontal above the strings and ready to play



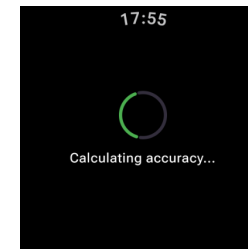
Countdown

Before execution



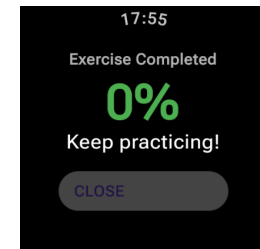
Execution

Gyroscope streams, phone records



Calculating

Phone runs onset detection

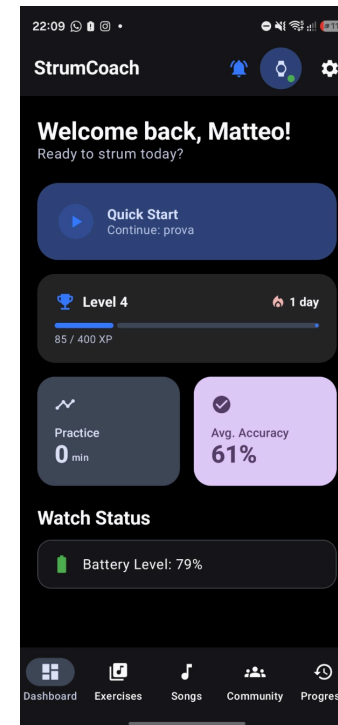


Result

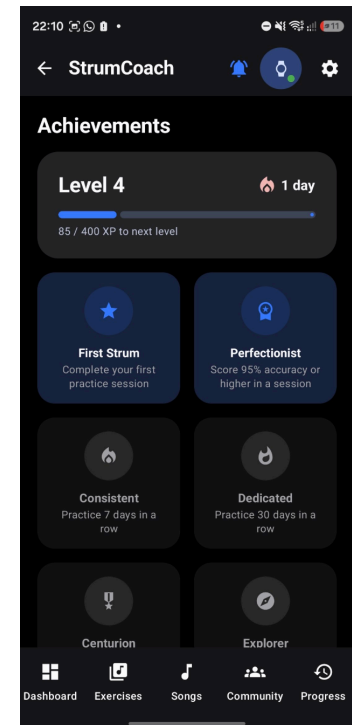
Accuracy and grade on the wrist

Gamification

- **XP:** every practice session earns 10 XP plus half its accuracy score
- **Levels:** level n costs $100 \cdot n$ XP, so progression slows on purpose
- **Streak:** consecutive calendar days with at least one session, read from session timestamps in local time
- **Badges:** eight achievements, each a predicate over the session history — surfaced on the session report the moment they unlock



Dashboard



Achievements

Personalization and reminders

- **Detection Sensitivity:** sets the Spectral Flux algorithm sensitivity
- **Haptic Intensity:** four different options
- **Daily Reminder:** a notification to remind the user to use the app, silent if a session was already recorded today
- **Onboarding:** a four-page tutorial on first launch, replayable from settings

Theme

Light	Dark	System
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Detection sensitivity

Low	Medium	High
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Haptics

Off	Low	Med	High
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Reminder

Morning	Afternoon	Evening
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Time to practice

Don't lose your 7-day streak.



Usability

Feedback lives on more than one channel, so a session can be run and read without watching a screen.

- **Haptic Beat:** the countdown is vibrated, so the start cue is felt rather than seen
- **Haptic Result:** a success waveform fires when the score arrives
- **Ambient Mode:** the summary stays legible when the display dims

A full session without looking at a screen

Felt

the countdown is vibrated on the wrist



Played

eyes stay on the fretboard, not on the phone



Felt again

a success waveform marks the end of the run



Read

the score lands on the watch, readable in ambient mode



Limitations and future work

Limitations

- **Not Streaming:** flux reprocesses the whole recording after execution, so feedback is post-session, not per stroke.
- **Weak Fallback:** the gyro-only path cannot separate a stroke from a return swing.
- **Fixed Thresholds:** tuned on the recordings available in development; the sensitivity setting shifts them, but does not learn.
- **Informal Validation:** manual tuning against a handful of real recordings, no controlled study across players or environments.

Future directions

- **Incremental Flux:** streaming the computation would give live, per-stroke feedback during practice.
- **Per-User Calibration:** a short exercise learning grip strength and typical strum energy instead of fixed constants.
- **ML Detector** Try to implement a ML algorithm to detect strums
- **Adaptive Difficulty:** using the accuracy trend to suggest the next exercise, rather than leaving the choice to the player.





Thank you for the attention!

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